

Characterization of polyoxyethylenes according to the number of hydroxy end groups by hydrophilic interaction chromatography at critical conditions for polyethylene glycol

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Abstract Polyoxyethylenes with different functionality and architecture can be separated according to the number of terminal hydroxy groups on polar stationary phases in acetone water mobile phases containing 90–97% acetone. The best results were obtained on a HILIC column. Typical samples, which can be analyzed by this technique, are polyethylene glycols, their mono- and dialkyl ethers, macromonomers, the fatty esters of PEG, and ethoxylated glycerol.

Keywords PEG · Ethoxylate · Functionality · Liquid chromatography under critical conditions · HILIC

Abbreviations

EO	Ethylene oxide
HILIC	Hydrophilic interaction liquid chromatography
LAC	Liquid adsorption chromatography
LCCC	Liquid chromatography at critical conditions
MALDI-TOF-MS	Matrix-assisted time-of-flight mass spectroscopy
PEG	Polyethylene glycol
PEG-MA	Polyethylene glycol monomethacrylate
PEG-MME	Polyethylene glycol monomethyl ether
PPG	Polypropylene glycol
SEC	Size exclusion chromatography

Introduction

Polyoxyethylenes are hydrophilic polyethers which have many applications in industrial manufacturing, medicines and personal care products [1–3]. Polyoxyethylenes with terminal hydroxy groups and a molar mass below 20,000 g/mol are typically referred to as polyethylene glycols (PEGs), samples with higher molar mass as polyethylene oxides. Polyoxyethylenes of different molar masses may have different physical properties but the same chemical properties.

PEGs and their derivatives are synthesized by anionic polymerization of ethylene oxide with various catalysts and initiators, resulting in products with different functionality. There are also stars with varying number of arms available.

Difunctional PEG (with two hydroxy groups) are obtained with water or a diol (e.g., di- or triethylene glycol). Monofunctional PEGs (with only one terminal hydroxy group) are synthesized using a monofunctional alcohol as initiator [4, 5]. Diethers of PEG [6] are obtained by deactivation of the growing chain by suitable coupling reactions. PEG derivatives with higher functionality can be obtained using multivalent alcohols (such as glycerol, trimethylol-propane, pentaerithrol, etc.). Other functionalities can be introduced by derivatization of the terminal hydroxy groups.

Polyoxyethylene is also used as hydrophilic block in the synthesis of amphiphilic surfactants that have applications in the pharmaceutical, cosmetic, and food industries. Fatty alcohol ethoxylates can only be synthesized by ethoxylation of fatty alcohols, while the fatty esters of PEG derivatives can be obtained by two different routes: ethoxylation of fatty acids, fatty esters, mono, di-, and triglycerides or esterification of the hydroxy groups PEG or other ethoxylates with fatty acids. Obviously, both routes will result in

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a mixture of the desired product (e.g., a monoester of PEG) with the unreacted starting materials and side products (fatty acids, PEG, and diesters).

Ethoxylated glycerides are well-known safe non-ionic surfactants [7–9] that have been used in the food processing, textile applications, and leather processing, pharmaceutical, cosmetic, and as agriculture ingredients. In cosmetics, they reduce the irritation potential of some aggressive surfactants. Other widely used ethoxylates of natural sugars are sorbitol ethoxylates and its esters known as polysorbates or tweens [10, 11].

PEGs are also used for the synthesis of polyurethanes [12]: polymer segments derived from PEG impart flexibility to the polyurethane for their applications as elastomeric fibers.

Another application is the synthesis of macromonomers [13–15] and crosslinkers for hydrogels: monoacrylates or methacrylates of PEG can be polymerized (or copolymerized with other monomers) to yield graft copolymers with a (meth)acrylate backbone and PEG grafts. Dimethacrylates are crosslinkers for the production of hydrogels, which can be used as solid electrolytes [16–18]. In the synthesis of soluble graft copolymers, a macromonomer must not contain dimethacrylates, as this would lead to crosslinked products.

Obviously, a full characterization of functional PEGs is crucial for many applications. This requires the determination of their molar mass distribution (MMD) and functionality type distribution. A rather difficult task is also the quantitative determination of other functionalities in trace amounts. As has been shown previously, this can be done by combinations of different chromatographic techniques (which will be discussed later on) and matrix-assisted time-of-flight mass spectroscopy (MALDI-TOF-MS) [1, 19–24]. The latter is an excellent tool in the identification of different functionalities, it is, however, rather problematic in the quantification of trace amounts, especially such of low molar mass.

In this paper, we have tried to find new approaches in HPLC, which would allow a separation of polyoxyethylenes with different functionalities under conditions, which also allow a reliable quantitative determination of impurities even in trace amounts.

Experimental

These investigations were performed using different instrumentation:

System A was an isocratic HPLC system Ultimate 3000, consisting of a pump ISO-3100A, column thermostat TCC-3000, autosampler WPS-3000SL, an RI detector RI 101, all from Dionex (Germerink, Germany). Mobile phases were

mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602 M (A. Paar, Graz, Austria).

System B was a gradient system Ultimate 3000, consisting of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, autosampler WPS-3000SL, all from Dionex (Germerink, Germany), an evaporative light scattering detector PL-ELS 2100 (Polymer Laboratories/Varian, Church Stretton, Shropshire, UK). All separations were performed at a flow rate of 0.5 ml/min. The mobile phases were mixed by the gradient pump, so their compositions are given in volume per cent.

On both systems, data acquisition and processing was performed using the software Chromeleon (Dionex, Germerink, Germany).

The following columns were used in this study:

- Spherisorb 5 μ Silica (Waters, Milford, MA, USA): plain silica; 250 $\times$ 4.0 mm; particle diameter: 5 μ m, pore size: 80 Å
- LiChrosorb Diol (Knauer, Berlin, Germany): silica-based diol phase, 250;4.6 mm, particle size: 5 μ m
- Nucleosil 100-5 OH 5 μ m (Macherey-Nagel, Dueren, Germany): silica-based diol phase; 250 $\times$ 4.6 mm; particle diameter: 5 μ m; pore size: 100 Å
- Discovery HS PEG (Supelco, Bellefonte, PA, USA): silica-based PEG phase; 250 $\times$ 4.6 mm; particle diameter: 5 μ m; pore size: 120 Å
- Synergi MAX RP (silica-based dodecyl phase; 250 $\times$ 4.6 mm; particle diameter = 4 μ m; nominal pore size = 80 Å, surface area: 475 m² g⁻¹; carbon load: 17%. from Phenomenex, Torrance, CA, USA) at a flow rate of 0.5 ml/min.
- Luna hydrophilic interaction liquid chromatography (HILIC) 5 μ m column (Phenomenex, Torrance, CA, USA; 250 $\times$ 4.6 mm; particle diameter = 5 μ m; pore size = 200 Å, carbon load: 5.7%)

The solvents (acetone and water, both HPLC grade) were purchased from Roth (Karlsruhe, Germany). PEGs and their mono- and dimethyl ethers were purchased from FLUKA (Buchs, Switzerland), macromonomers from Sigma-Aldrich.

Ethoxylates of fatty acids, mono- and triglycerides were provided by the Institute of Heavy Organic Synthesis (ICSO), Kedzierzyn-Kozle (Poland).

Results and discussion

Functional polymers and oligomers can be analyzed by HPLC in different modes, which are characterized by the so-called interaction parameter [25]. In size exclusion

chromatography (SEC), separation is driven by the entropy change, and the interaction parameter is strongly negative. Polymers are separated according to their size. Large molecules, which have no access to the pores of the stationary phase, are eluted earlier, while smaller ones spend more time in the pores and are eluted later. SEC is in widespread use for the determination of the MMD of polymers [1]. No information is, however, obtained on functionality. Liquid adsorption chromatography (LAC) is typically governed by the enthalpy change, and the interaction parameter is positive. Retention increases with the number of groups, which may interact with the stationary phase. In liquid chromatography at critical conditions (LCCC) enthalpic and entropic effects compensate each other, and retention becomes independent on molar mass.

Liquid adsorption chromatography

In LAC, the separation order depends on the nature of the stationary and mobile phase. In normal phase systems (where the stationary phase is more polar than the mobile phase), retention increases with the number of polar groups. If the end groups can interact with the stationary phase, their interaction parameter (the end-group parameter q [26, 27]) must also be taken into account.

In the case of PEG derivatives, this means, that retention depends on these parameters, on the number of ethylene oxide (EO) units and the number of hydroxy groups. As is shown in Fig. 1, the individual oligomers of PEG derivatives can be resolved on a silica column in 90–95% acetone, and retention increases in the order polyethylene glycol dimethyl ether < polyethylene glycol monomethyl ether (PEG-MME) < PEG. In reversed phase systems, retention depends on the number of EO units and alkyl end groups [26, 27]. As has been shown previously, a full

resolution of all oligomers in a mixture of PEG and PEG-MME can be achieved under appropriate conditions [28]. At a special mobile phase, composition and temperature the second homologous series can be shifted in a way that all peaks elute exactly between those of the main series. This is, however, not possible for more complex samples, which contain more than two polymer homologous series. In this case, the influence of the main chain on retention has to be suppressed.

Liquid chromatography at critical conditions

In a special mobile phase composition (and temperature), the polymer chain becomes “chromatographically invisible” [29–32]. This is called the critical adsorption point (CAP), where entropic and enthalpic contributions compensate each other, and the interaction parameter equals zero. At the CAP for the EO unit, all polyethylene oxide chains elute at same elution volume regardless of their length. This separation mode allows a separation according to the end groups or other blocks.

For functional polymers and block copolymers, the interaction parameter of the other block (or the end-group parameter) determines the elution behavior. If the polymer contains another block, the interaction parameter of which is negative, the copolymer elutes in SEC mode, i.e., earlier than the homopolymer at the CAP, which allows a determination of the MMD of this block.

If a polymer contains one end group (or block), which is adsorbed at the CAP of the repeat unit, all oligomers elute in a sharp peak (independent of molar mass) at an elution volume corresponding to that of the corresponding end group, and are separated from the non-functional oligomers, which also elute as a narrow peak at the void volume of the column [33, 34]. Under such conditions, polymer homologous series with different functionality can be separated

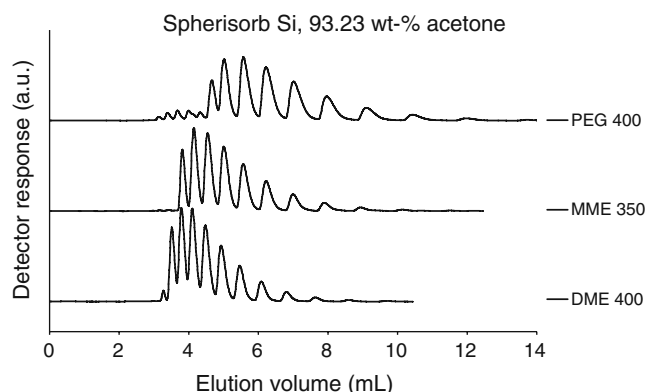


Fig. 1 Separation of PEGs with different end groups on Spherisorb Si in 93 wt.% acetone

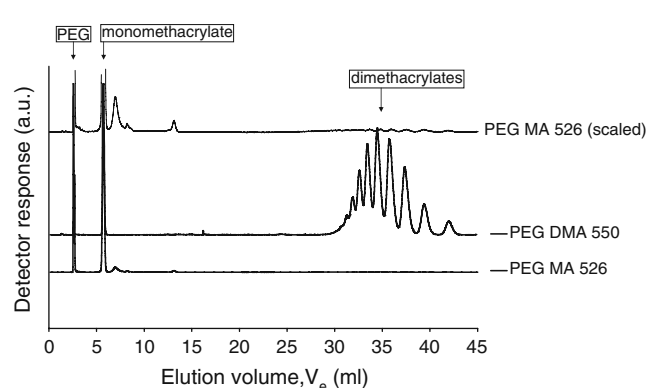
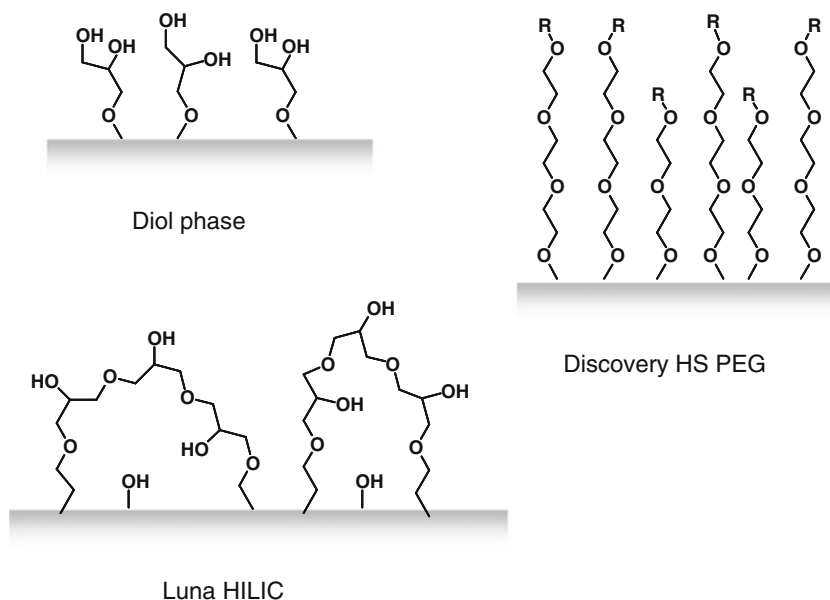


Fig. 2 Separation of PEG and a PEG methacrylate on Synergi MAX RP at the CAP for EO in 32 vol.% acetone (detection: ELSD)

Fig. 3 Stationary phases modified with polar groups



from each other. Monofunctional molecules and A–B diblocks (with an adsorbing block A) elute at the CAP for B as narrow peaks, and so do B–A–B triblocks.

On a reversed phase column, this can even be achieved, if the end groups are not very different from the repeat unit, as is the case with PEG and PEG–MME: under critical conditions for PEG on a reversed phase column in acetonitrile–water, acetone–water or methanol–water mobile phases both series elute as narrow peaks, which are well separated from each other [4, 19, 20].

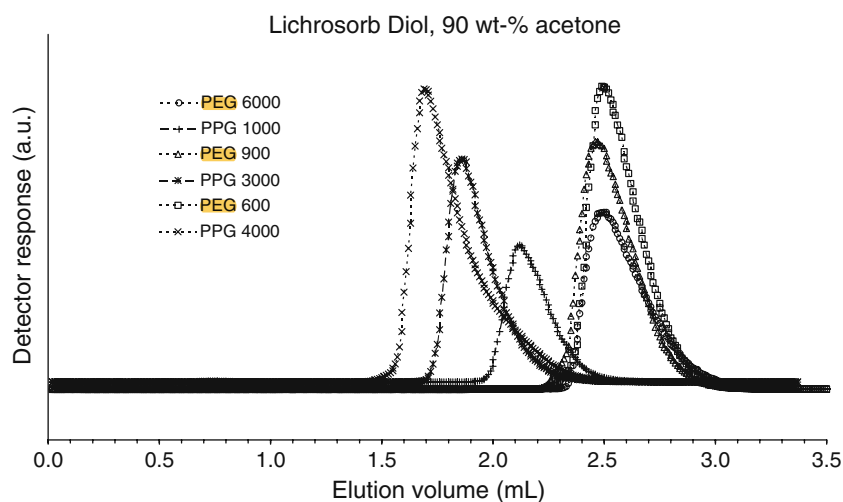
A very different elution behavior is, however, observed with A–B–A triblocks (or difunctional chains) [30, 35]. The probability of the second end group (or block) to be adsorbed depends on the size of the block in between: even at the CAP for B, the retention of A–B–A triblocks is not

independent on molar mass: it decreases with increasing length of the block B. This means, that the individual oligomers of this series are separated in SEC order, but far behind the void volume [35–37], as is illustrated by Fig. 2.

This can be utilized for the separation of diesters, but it is a serious problem, if small amounts of diesters have to be detected, which would fall beyond the detection limit. A reliable quantification would only be possible, if they elute as a reasonably narrow peak. Evidently, the macromonomer (polyethylene glycol monomethacrylate 526) contains some dimethacrylates, but their quantitative determination will be difficult.

Even more complicated is the situation with amphiphilic stars, such as the higher esters in polysorbates. A separation of such samples according to the number of hydrophobic

Fig. 4 Separation of PEGs and PPGs with different molar mass on **Lichrosorb Diol** in 90 wt.% acetone



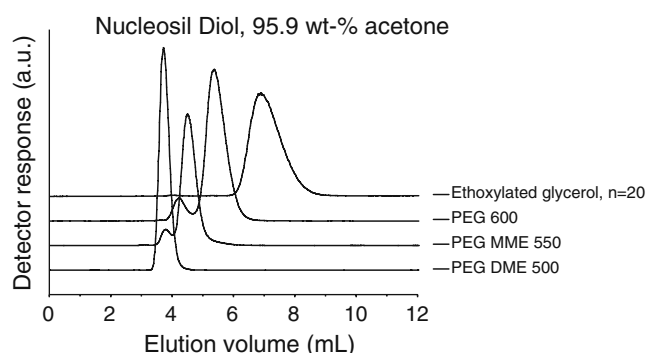


Fig. 5 Separation of PEG derivatives with different number of hydroxy end groups on Nucleosil OH in 95 wt.% acetone

groups at the CAP for PEG will result in very complex chromatograms.

Obviously, a separation of different polymer homologous series in polyoxyethylenes according to their functionality can be problematic on reversed phase columns.

In another communication [38], we have tried to suppress the molar mass dependence of retention of the difunctionals on a reversed phase column using a gradient along the critical adsorption line, which exists in mobile phases consisting of acetone, methanol, and water [39].

One may, however, speculate that it might also be possible to achieve such a separation on normal phase systems. On plain silica, this does not work, as can be concluded from Fig. 1: evidently, the interaction parameter of the EO unit and the end group parameter of the hydroxy group are too similar. If a CAP for EO could be found on a silica column, the hydroxy selectivity would be by far too low. There are, however, other polar stationary phases, which could be used for such a separation: diol phases, a PEG-modified phase (Discovery HS PEG), and a phase modified with polyglycidol (Luna HILIC), the structure of which are shown schematically in Fig. 3.

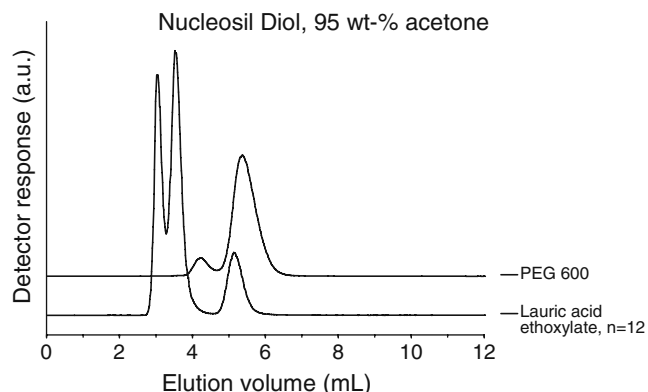


Fig. 6 Separation of PEG 600 and a fatty acid ethoxylate on Nucleosil OH in 95 wt.% acetone

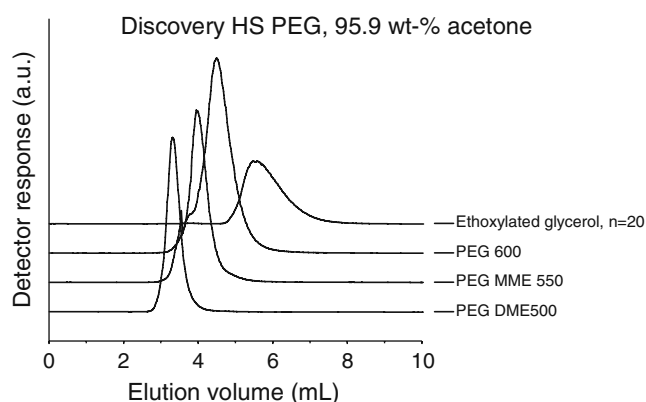


Fig. 7 Separation of PEG derivatives with different number of hydroxy end groups on Discovery HS PEG in 95.9 wt.% acetone

The purpose of the present work was to find a system, which should allow a separation of polyoxyethylenes according to the number of hydroxy groups.

As has already been described previously [40], a CAP for PEG could be found on a diol phase in about 80 wt.% acetone–water. Under such conditions, polypropylene glycol (PPG) eluted in SEC mode. **The same behavior is also observed on diol phases in 90% acetone (Fig. 4).**

We have thus analyzed polyoxyethylenes with different functionality on such a column and found, that they were really separated according to the number of hydroxy groups (Fig. 5).

In the separation of fatty acid ethoxylates, the same retention order was observed: the first peak corresponds to the diester, the second one to the monoester, and the third one to PEG (Fig. 6).

This encouraged us to try also the other columns: on the Discovery column, the same elution order was observed, but with much poorer efficiency (Fig. 7).

The best results were obtained with the Luna HILIC column (which is typically used for other purposes [41–44]): as can be seen in Fig. 8, polyoxyethylenes with

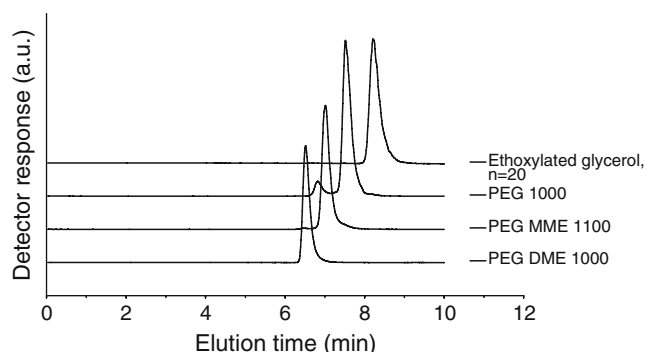


Fig. 8 Separation of PEG derivatives with different number of hydroxy end groups on Luna HILIC in 97 wt.% acetone

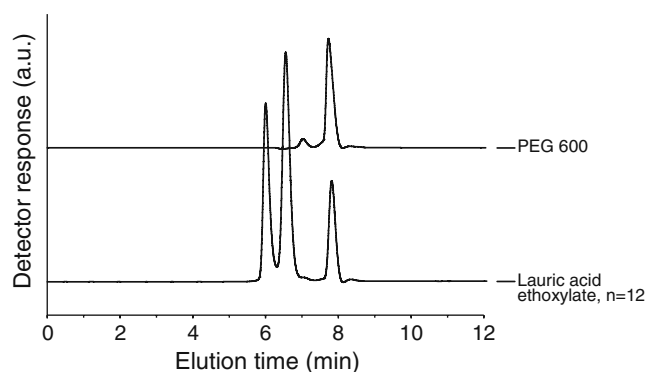


Fig. 9 Separation of PEG 600 and a fatty acid ethoxylate on Luna HILIC in 97 wt.% acetone

different number of hydroxy groups could be very well separated from each other.

A baseline separation of all peaks was also achieved on this column with fatty ester ethoxylates (Fig. 9). In another communication, we have applied this concept to a more complex kind of surfactants: polysorbates could be separated by two-dimensional comprehensive liquid chromatography with hydrophilic interaction chromatography and LCCC on a reversed phase column as first and second dimension [45]. The individual fractions could be identified by MALDI-TOF-MS (Abrar et al. in preparation).

The question was now, whether these conditions really corresponded to the CAP for PEG. This is indeed the case, as can be seen in Fig. 10. The slight deviation from critical behavior in 97% acetone is no problem for samples with similar molar mass. On the other hand, the shift in retention between the different series is largest in this mobile phase.

Under the same conditions, other polyethers (PPG and poly(tetrahydrofuran)) eluted in SEC mode (Fig. 10). The lines obtained with these polymers coincide.

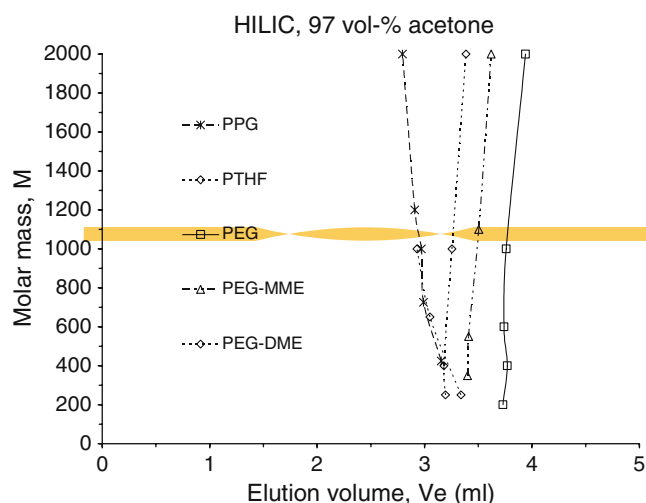


Fig. 10 Elution volumes of different polymers on Luna HILIC in 97 wt.% acetone

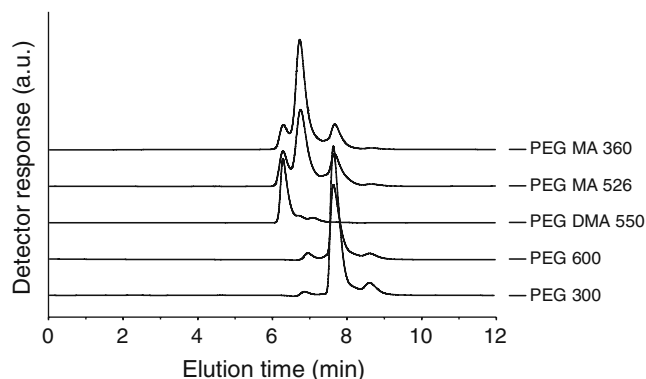


Fig. 11 Separation of PEGs and PEG methacrylates on Luna HILIC in 97 wt.% acetone

A quite interesting result could be obtained in the analysis of macromonomers: as can be seen in Fig. 11, both macromonomers contain PEG (which is not dramatic), but also considerable amounts of dimethacrylates, which will lead to crosslinking in the polymerization, and thus to the formation of hydrogels instead of soluble graft copolymers. The determination of small amounts of dimethacrylates in macromonomers is the subject of another communication [38].

Another application is the analysis of ethoxylated glycerides: as can be seen in Fig. 12, an ethoxylated monoglyceride contained considerable amounts of ethoxylated diglyceride and glycerol as well as traces of PEG (resulting from chain transfer to water), while a sample of ethoxylated rapeseed oil did not contain these side products.

Conclusions

The results described above suggest a somewhat different mechanism compared to the typical one in LCCC: if the separation would be based on adsorption, difunctionals should be resolved as predicted by the theory [30, 35].

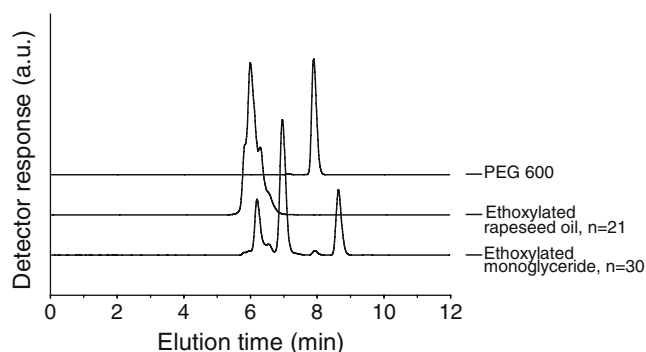


Fig. 12 Separation of PEG 600 and two ethoxylates on Luna HILIC in 97 wt.% acetone

Evidently, the chromatograms obtained in this study do not show this behavior, which leads to the assumption, that the separation on these columns should rather be based on partitioning into the aqueous (or at least water-rich) layer at the surface of the stationary phase. Further studies shall show the thermodynamic background of these separations.

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